DO NOW — RECALL

Floating-point representation and normalisation

Name: _____ Class: _____ Date: _____

1. Write −2 in 8-bit two's complement.

2. What is unsigned 00100000 after a logical right shift by one?

3. Convert binary 0.1_2 to decimal.

4. Calculate 0.5×2^3.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

11111110.

2. Model answer

00010000.

3. Model answer

0.5.

4. Model answer

4.

Vocabulary — know these terms

mantissa — The signed significant part of a floating-point number.

exponent — The power of two scaling the mantissa.

normalised — A two's-complement fractional mantissa with different first two bits.

precision — Number of significant bits represented.

range — Spread of magnitudes that can be represented.

NEW LEARNING

Read, annotate and explain

Why is 0.0110 not normalised?

For a positive two's-complement fractional mantissa, a normalised value begins with 01, not 00.

Shift and compensate

Shift 0.0110 left one place to 0.1100. Decrease the exponent from 4 to 3 to compensate.

Verify the value

0.0110_2 is 0.375; $0.375 \times 16 = 6$. 0.1100_2 is 0.75; $0.75 \times 8 = 6$.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Normalise positive $0.0011_2 \times 2^5$.

First shift

0.0011 becomes 0.0110; reduce exponent 5 → 4. It still begins 00.

Second shift

0.0110 becomes 0.1100; reduce exponent 4 → 3. The first two bits are now 01.

Check

Both represent 6 in decimal, so normalisation preserved the value.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Identify mantissa and exponent in $0.1100_2 \times 2^3$.

Question 2

Normalise positive $0.0010_2 \times 2^5$.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Why can normalisation improve precision without increasing the number of mantissa bits?

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Shifting 0.0110×2^4 left to 0.1100×2^4 preserves its value.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.